CaaS Platforms

Now that we've had a brief overview of our options if we choose to build on top of IaaS, let's turn our attention to the Container as a Service (CaaS) space.

As we've already discussed, CaaS offerings let you hit the ground running faster. They provide a managed service, where your starting point is a platform capable of running Dockerized applications.

In this area, *Kubernetes is king*. It is the predominant orchestration service offered by providers. This means that, once your Kubernetes workload cluster is up and running, you interact with it via the `kubectl` command to deploy, update, or scale your app. Similarly, your config manifests that describe the services you want running and how they should be connected will need to be in Kubernetes format.

Each offering is at a slightly different place in terms of features and maturity, so let's review the major players.

Amazon Elastic Container Service

The odd one out in this list, Amazon Elastic Container Service (Amazon ECS),[15] sports Amazon's own container orchestration layer, rather than Kubernetes.

The AWS *way*—ECS being no exception—is to provide the building blocks to architect your cloud infrastructure and apps from the ground up. This gives you very fine-grained control, but the trade-off is that you are exposed to additional complexity and have more work to hook things together. Most people either love it or hate it.

The service can be managed through the AWS Management Console[16] or via the ECS CLI,[17] both of which let you provision the infrastructure needed. Additionally, CloudFormation[18]—Amazon's proprietary infrastructure provisioning language—can also be used.

In ECS, you describe your application's containers using config files known as task definitions[19] in JSON format. It also provides compatibility with

15. https://aws.amazon.com/ecs/
16. https://aws.amazon.com/console/
17. https://docs.aws.amazon.com/AmazonECS/latest/developerguide/ECS_CLI.html
18. https://aws.amazon.com/cloudformation/
19. https://docs.aws.amazon.com/AmazonECS/latest/developerguide/task_definitions.html